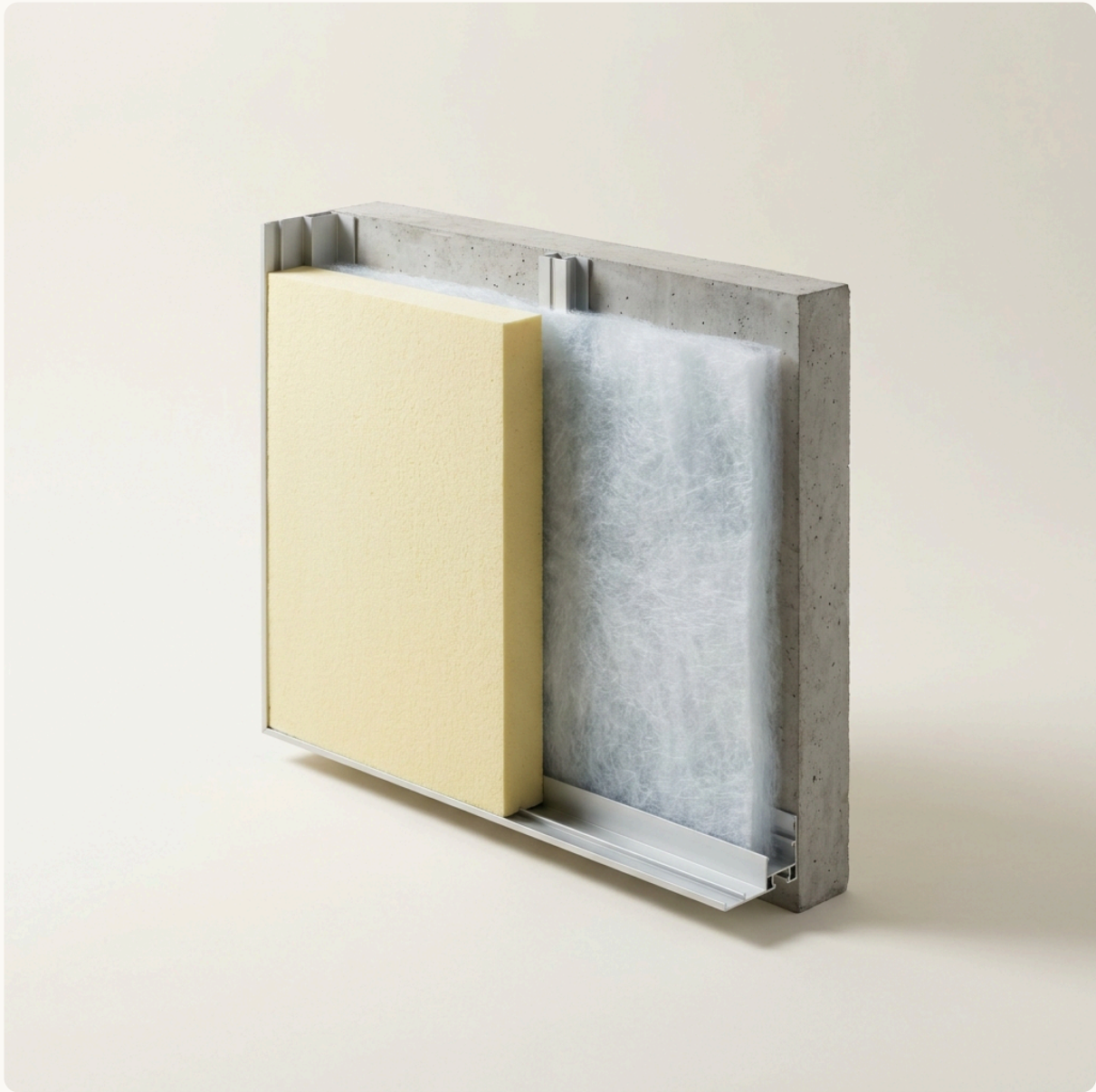


LEDGER · 2026-08-31 · MATERIALS SCIENCE / THERMAL INSULATION

Polyimide-Reinforced Silica Aerogel Blankets via Ambient Pressure Drying

PRODUCT



A product dossier of The Absence Audit. The current audit flags this concept — Margin fails the operating-cost check — and the full disclosure ships inside the dossier.

Incumbent benchmark

Aspen Aerogels Pyrogel XTE

Entry application

High-temperature industrial piping and processing vessels in petrochemical and refining sectors

Measured advantage	Thermal conductivity of 0.018 W/mK vs Pyrogel XTE's 0.014 W/mK (28% performance regression) and maximum temperature rating of 500°C vs 650°C
Time to first revenue	12 months
Gross margin at parity	56.4%
Startup CapEx	\$350,000
Payback from first sale	2.2 months
Capital productivity	13.98x
Regulatory risk	Moderate — qualification costs reported (\$150,000 estimated)

BUY THE FULL DOSSIER

Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

[Buy the full dossier — \\$499.00 →](#)

Assessed 2026-08-31 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

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Track record

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